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#include <stdio.h>

struct Process {
    int pid, at, bt;
    int ct, tat, wt, rt;
    int remaining;
    int started;
};

int main() {
    int n;
    printf("Enter number of processes: ");
    scanf("%d", &n);

    struct Process p[n];

    for(int i = 0; i < n; i++) {
        p[i].pid = i + 1;
        printf("\nProcess %d\n", p[i].pid);
        printf("Arrival Time: ");
        scanf("%d", &p[i].at);
        printf("Burst Time: ");
        scanf("%d", &p[i].bt);
        p[i].remaining = p[i].bt;
        p[i].started = 0;
    }

    int time = 0, completed = 0;
    float avg_tat = 0, avg_wt = 0, avg_rt = 0;

    while(completed < n) {
        int min = -1;

        for(int i = 0; i < n; i++) {
            if(p[i].at <= time && p[i].remaining > 0) {
                if(min == -1 || p[i].remaining < p[min].remaining)
                    min = i;
            }
        }

        if(min == -1) {
            time++;
        } else {

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        if(p[min].started == 0) {
            p[min].rt = time - p[min].at;
            p[min].started = 1;
        }

        p[min].remaining--;
        time++;

        if(p[min].remaining == 0) {
            p[min].ct = time;
            p[min].tat = p[min].ct - p[min].at;
            p[min].wt = p[min].tat - p[min].bt;
            completed++;

            avg_tat += p[min].tat;
            avg_wt += p[min].wt;
            avg_rt += p[min].rt;
        }
    }
}

printf("\nPID\tAT\tBT\tCT\tTAT\tWT\tRT\n");
for(int i = 0; i < n; i++) {
    printf("%d\t%d\t%d\t%d\t%d\t%d\t%d\n",
        p[i].pid, p[i].at, p[i].bt,
        p[i].ct, p[i].tat, p[i].wt, p[i].rt);
}

printf("\nAverage Turnaround Time = %.2f", avg_tat/n);
printf("\nAverage Waiting Time = %.2f", avg_wt/n);
printf("\nAverage Response Time = %.2f", avg_rt/n);

return 0;
}

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C:\Users\Admin\Desktop\1 x + v
Enter number of processes: 3

Process 1
Arrival Time: 2
Burst Time: 5

Process 2
Arrival Time: 3
Burst Time: 2

Process 3
Arrival Time: 1
Burst Time: 4

PID    AT    BT    CT    TAT    WT    RT
1       2     5     12    10     5     5
2       3     2     5     2     0     0
3       1     4     7     6     2     0

Average Turnaround Time = 6.00
Average Waiting Time = 2.33
Average Response Time = 1.67
Process returned 0 (0x0)  execution time : 17.113 s
Press any key to continue.
```

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SJF(P)